Robust Bayesian estimation of nonlinear parameters on SE(3) Lie group from noisy measurments

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Abstract—Sensors are physical devices that take measurements of a source from a specific pose. In general the pose constitutes of three parameters for the position and another three for the heading. In many scenarios the geometric relationship between sensor and source is precisely known. However, in the field of autonomous robot navigation, neither the robot (sensor) pose nor the environment (source) is known, but in principle can be determined by measurements. The problem of self localization from noisy measurements is known as SLAM. Possible sensors are laser range finders (3D data) or cameras (2D data measurements). Interestingly, the quality of the pose estimates depends on its particular parametrization. We show that in the case of merging 3D range data, the full exponential Lie Cartan coordinates of 1.st kind provide the best unbiased estimates when assuming a Gaussian noise model on the measurements. Further, we show that in the case of projective 2D camera measurements, a joint posterior estimate of camera pose and environment is computationally not advantageous. Instead, a marginalized posterior over the environment should be considered. The marginalization over the non-gaussian variables has to be performed carefully using a Saddlepoint approximation method.

I. Introduction

The basic challange of automomous robotic exploration is to safely interact with natural environments. Sensors, such as cameras, laser range finders or ultra-sound constitute the essential input. In order to plan and execute a route towards a target, a three dimensional representation of the environment will be needed. Such a map can be given a priori or has to be built along the way. In either way, the robot has to localize itself within the environment.

Traditionally, video cameras were thought to be a proper input for a guidance system. Years of computer vision research has led to numerous approaches of how to extract useful information from an array of numbers, that is indeed the internal computer representation of a camera image. One key interest in this research field, is the identification of objects. Another one is the inference of the 3 dimensional geometry of the environment. The general problem to infer 3 dimensional geometry from 2 dimensional motion is described as structure form motion (SFM). Robustness and reliability requirements have led to the development of a statictical framework to address this problem. Statistical sampling techniques, such as Least Median Squares (LMedS) , RANSAC and maximum likelihood (ML) or maximum a posteriori (MAP) estimators are used in this contex.

In robotics research, the problem of navigating through unkown environments is addressed in a slightly different formulation as simultaneous localization and mapping (SLAM). Here, the solution techniques are based on recursive probabilistic inference. Noteably, the use of particle filter (PF) techniques for the use of probability representations dominates this field.

Both in recursive methods such as in SLAM and in the bundle-ajustment techniques used in computer vision the common problem is to determine the robot heading, or camera pose. Once the camera pose(s) is known, triangulation in the case of camera observation or simply fusion of laser range finder maps readily yields the 3 dimensional map. In general, 6 degrees of freeedom (DOF) have to be considered for the pose estimate. Uncertainty in the pose estimate can produce significant distortion in the 3D map. For most accurate pose determination, ML/MAP estimation on the whole data set has proven to be the best approach. This techniques is called Bundle-Adjustment in computer vision and is intrinsically non-recursive and rather computationally expensive since it may involve large data sets.

In this paper we address the problem of highly accurate pose estimation. We will formulate a parametric uncertainty representation for the camera pose and develop an efficient estimation scheme. We will use the notion of the special eucldian group $SE(3)$, as a framework for inference. The main focus will be on the application towards the bundle-adjustment procedure. The role of parametrization: It was observed that proper parameterization of the camera pose plays a crucial role. Euler angle parametrization of the orientational component of the camera pose proved to be the least numerically stable approach. Global quaternion or Rodrigues parametrization greatly improve the stability of least squares minimization algorithms. Up to date techniques use incremental, local, Rodrigues parametrization [8]. Whereas the focus for bundle-adjustment lies on the performance of non-linear

minimization algorithms, recursive methods deal with finite probability parametrizations to capture the error model. To our knowledge, the role of error models in parameter estimation over a non-euclidian manifold such as $SO(3)$ and $SE(3)$ hav been rarely addressed. In [12] the notion of compositional noise was adopted.

II. Probabilistic framework for the SLAM problem

The simultaneous localization and mapping (SLAM) problem is adequately formulated in a probabilistic framework. Solving SLAM is finding the joint probability distribution of the camara pose f_t and a set of 3D landmarks $\{\mathbf{x}^j\}_t$ representing the knowledge about the environment at present time step t

$$P(f_t, \{\mathbf{x}^j\}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t}) \tag{1}$$

given all past and present observations $\mathbf{z}_{1:t}$ and if available the history of internal odometry data $\mathbf{u}_{1:t}$. The 3D landmarks $\{\mathbf{x}^j\}$ and the parameterized camera pose f_t are defined with respect to a global world frame W. Also, we have assumed that the association (correspondence) problem between observed feature z^j and landmark $\mathbf{x}^j$ is solved.

A. Recursive filter versus bundle adjustment

Using Bayes theorem a solution to (1) can be derived recursively,

$$\begin{aligned} P(f_t, \{\mathbf{x}^j\}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t}) \propto P(\mathbf{z}_t | f_t, \{\mathbf{x}^j\}_t) \times \\ \iint P(f_t | f_{t-1}, \mathbf{u}_t) P(\{\mathbf{x}^j\}_t | \{\mathbf{x}^j\}_{t-1}) \times \\ P(f_{t-1}, \{\mathbf{x}^j\}_{t-1} | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) \, df_{t-1} \prod_j d\mathbf{x}^j_{t-1}, \end{aligned} \tag{2}$$

where the proportionality only neglects the measurement error distribution. Eqn. (2) is exact for general probability distributions and states that we can make optimal use of all the measurements in a serial (recursive) fashion; We keep only knowledge of the present camera pose f_t and the 3D environment $\{\mathbf{x}^j\}_t$, which is derived by weighing the knowledge about the robots pose and the 3D environment at a previous time step against the current observation. Practically, Eqn. (2) is approximated by a Kalman filter (KF) technique, which assumes that all probabilities are sufficiently described by at most second order moments.

In bundle adjustment, the full robot path $f_{1:t}$ and the 3D environment is inferred simultaneously. At first glance, it seems that valuable information is lost in the recursive procedure (2), since we only retain the current pose. However, marginalizing the bundle adjustment result over all the past camera poses $f_{1:t-1}$, will yield the recursive solution (2),

$$\begin{aligned} P(f_t, \{\mathbf{x}^j\}_t | z_{1:t}, u_{1:t}) = \int P(f_{1:t}, \{\mathbf{x}^j\}_t | z_{1:t}, u_{1:t}) \, df_{1:t-1} \\ \propto P(z_t | f_t, \{\mathbf{x}^j\}_t) \int df_{t-1} \, P(f_t | f_{t-1}, u_t) \times \\ P(z_{t-1} | f_{t-1}, \{\mathbf{x}\}_t) \int df_{t-2} \, P(f_{t-1} | f_{t-2}, u_{t-1}) \times \dots \\ \int df_1 \, P(f_1, \{\mathbf{x}^j\}_t | z_1, u_1). \end{aligned} \tag{3}$$

Expressions (2) and (3) are equivalent if we assume a stationary 3D environment,

$$p(\{\mathbf{x}^j\}_t | \{\mathbf{x}^j\}_{t-1}) = \prod_j \delta(\mathbf{x}^j_t - \mathbf{x}^j_{t-1}). \tag{4}$$

Though, Eqns (2,3) are exact, from a practical standpoint we have to be concerned with the numerical representation of probability distributions. Mostly used in the SLAM problem context are ensemble representations with Eqn. (2) realized by particle filters (PF). Although, various KF implementations for the joint probabilities have been tested, a gaussian probabilty representation has been found computationally insufficient [?]. A simple recursive KF implementation of (2) will result in exceedingly large covariance matrices, therefore rendering it impractical for real applications with an increasing number of landmarks. Also, it has been reported that KF implementations result in underestimating the robot pose uncertainties, thus causing serious problems for loop-closing.

III. Parametric uncertainty represention for camera pose

As has been shown for a wide class of statisitcal processes, a finite parametric representation for the distribution is not available. In [?] it has been argued that at most we can expect to find a sub-optimal parametric representation. The simplest thereof is a system of gaussian random variables, with the mean μ called the statevector and Σ the covariance matrix.

Notation: vectors are denoted with bold face characters $\mathbf{x}, \mathbf{T}, \dots$ matrices are given as sans serif characters R, Σ.

A. Motivational experiment

Suppose we have a global map consisting of N 3D landmarks $\{\mathbf{x}^j\}$ with reference to the world frame. A robot equipped with a laser range scanner identifies the 3D landmarks in its particular camera frame $f_0 \in SE(3)$[1]. In general we only have knowledge of the identified 3D landmarks $\{\mathbf{x}^j\}$ in the global map and their corresponding range measurements in the yet to be determined camera frame f_0. The range measurements are essentially the identified 3D landmarks transformed into the camera frame $\{\mathbf{p}^j\}$. Then, our task is to recover the camera pose (camera localization) f_0, using only the matched sets of

[1] $SE(3) \cong SO(3) \times \mathbb{R}^3$ stands for the Special Euclidean group of rigid body motions in $\mathbb{R}^3$, see Appendix A

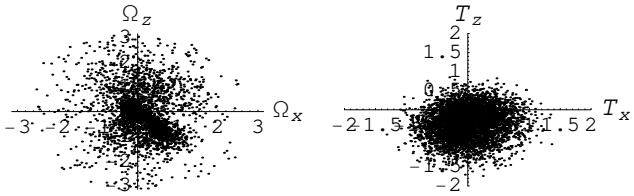

Fig. 1. Projected sample distribution for rotational components $\Delta\mathbf{\Omega} = \mathbf{\Omega} - \mathbf{\Omega}_{f_0}$ on the left side, and the translational coordinates $\Delta\mathbf{T} = \mathbf{T} - \mathbf{T}_{f_0}$ on the right. It is clearly seen that the translational estimates are biased.

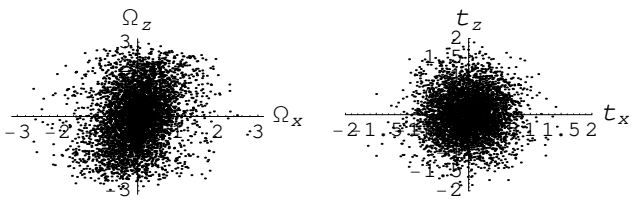

Fig. 2. The same sample distribution as in Fig.1 is plotted in SE(3) exponential coordinates, given by $\ln(f_0^{-1} \cdot f) = [\Delta\hat{\mathbf{\Omega}}, \mathbf{t}]$, the distribution is less biased since it is centered around the origin and seems to show gaussian behavior, see further sections and App. A.

3D points. In other words, we use the following Likelihood estimator,

$$f_0 = \arg \min_{f \in SE(3)} \sum_j^N \left| \mathsf{R}(\mathbf{\Omega}_f)\mathbf{x}^j + \mathbf{T}_f - \mathbf{p}^j \right|^2 . \quad (5)$$

Eqn. (5) can be solved exactly, the solution may not be unique for points on degenerate surfaces. In the center of "mass" coordinates $\mathbf{x}_c$ with $\mathbf{x} = \tilde{\mathbf{x}} + \mathbf{x}_c$, we can decouple the rotational and translational "modes", then the solution is

$$\mathbf{\Omega}_{f_0} = \arg \min_{\mathbf{\Omega}_f \in SO(3)} \sum_j^N \left| \mathsf{R}(\mathbf{\Omega}_f)\tilde{\mathbf{x}}^j - \tilde{\mathbf{p}}^j \right|^2 . \quad (6)$$

We solve for the optimal rotation $\mathsf{R}(\mathbf{\Omega}_{f_0}) \in SO(3)$, using Horn's quaternion method [13], [14]. Once, having found the optimal rotation it is straightforward to solve for $\mathbf{T}_{f_0} = \mathbf{p}_c - \mathbf{R}(\mathbf{\Omega}_{f_0})\mathbf{x}_c$. We call f_0 the zero noise estimate.

In any real scenario, we cannot assume impeccable knowledge of the 3D map nor perfect noise free measurements. For simplicity we model our inaccurate knowledge of the global map as additive gaussian noise with constant variance on the 3D landmarks $\mathbf{x}^j \sim \mathcal{N}(\bar{\mathbf{x}}^j, \sigma)$. We initialize our experiment with a critical number $N = 3$ points $\mathbf{x}^j$ randomly distributed in the cube $[-1, 1]$ around the origin. The camera is located outside the cube, $T_{f_0} = [0, 0, -10]$ and its view directed towards the center. The corresponding points $\mathbf{p}^j$ are computed. We sample the estimated camera pose parameter distribution, Fig. 1,2, by corrupting the original landmarks $\mathbf{x}^j$ with gaussian noise of zero mean and variance $\sigma = 1.0$.

Interestingly the estimated parameter distribution shows strong bias if we choose the minimal parametrization $[\mathbf{\Omega}, \mathbf{T}]_{L_2}$ as a conjunction[2] of Rodrigues vector and translational vector. For an unbiased estimate we expect the mean of the parameter sample distribution around the zero noise estimate $[\mathbf{\Omega}_{f_0}, \mathbf{T}_{f_0}]_{L_2}$. This is clearly not the case in Fig.1. However, if we choose to parameterize the camera pose by full exponential coordinates $[\mathbf{\Omega}, \mathbf{t}]_{L_1}$, we obtain an almost unbiased estimate, Fig.2. Some insight can be gained by the following argument: If we assume an unbiased distribution in exponential coordinates $< [\mathbf{\Omega}, \mathbf{t}]_{L_1} >= 0$ where correlations between rotational and translational components exist $< \Omega_\alpha t_\beta > \neq 0$, then due to the algebraic connection (51) between exponential coordinates and translational vector the estimate for the translational vector $\mathbf{T}$ will be biased $< \mathbf{T} > = < \mathsf{S}(\mathbf{\Omega})\mathbf{t} > \neq 0$.

B. Camera Pose in the $SE(3)$ Lie group context

In the context of continuous groups, exponential coordinates play an essential role. In this section we will briefly introduce the notion of groups to represent camera frames. More details can be found in the AppendixA as well as in abundant literature [?].

The Rodrigues vector $\mathbf{\Omega}$ constitutes the exponential parametrization of a rotation matrix $\mathsf{R}(\mathbf{\Omega})$. Together with the translation vector $\mathbf{T}$, we obtain a 6 parameter minimal representation of a frame. Intuitively, we can visualize a frame using the picture of a trihedron with 3 degrees of freedom (DOF) for the origin and another 3 DOF for the orientation. The set of all possible frames form the $SE(3)$ group, where we adopt the following notation for its elements

$$f \cong (\mathsf{R}(\mathbf{\Omega}_f), \mathbf{T}_f) \in SE(3). \quad (7)$$

The terms rigid body motion, frame and camera pose can be used synonymously. In order to see how f encodes the rigid body motion, frame or camera pose information, we define another important operation (action) on 3D points $\mathbf{x}$. A point $\mathbf{x}$ as well as a camera frame $f = (\mathsf{R}_f, \mathbf{T}_f)$ is defined with respect to a global world frame W, then a change to the camera frame will transform the coordinates of $\mathbf{x}$ according to

$$\mathbf{x}_f = f \star \mathbf{x} = \mathbf{R}_f \, \mathbf{x} + \mathbf{T}_f. \quad (8)$$

Vice versa, we could imagine that a set of 3D points $\mathbf{x}_f$ are rigidly attached to a frame f, then we speak of a rigid-body motion if time-dependent transformation parameters $f^{-1}(t) \cong (\mathsf{R}_{f(t)}, \mathbf{T}_{f(t)})$ can be found such that they explain the dynamical evolution of those points seen in the world frame, $\mathbf{x}(t) = f^{-1}(t) \star \mathbf{x}_f$.

[2]we denote the conjunction of real valued parameters according to MATLAB notation by $[a, b, \ldots]$. The subscript L_2 pertains to the notion of the coordinate system, details are in the Appendix. We note that the conjunction of various quantities doesn't constitute a proper vector.

The operation between group elements is the composition $g = h \cdot f$, here

$$(\mathsf{R}(\mathbf{\Omega}_h), \mathbf{T}_h) \cdot (\mathsf{R}(\mathbf{\Omega}_f), \mathbf{T}_f) = (\mathsf{R}(\mathbf{\Omega}_h)\mathsf{R}(\mathbf{\Omega}_f), \mathsf{R}(\mathbf{\Omega}_h)\mathbf{T}_f + \mathbf{T}_h). \tag{9}$$

The nature of this composition is a semi-direct product. Rotations only compose with rotations but translations also interact with rotations. Thus, the composition is non-commutative since finite rotations are not interchangeable with each other and with translations. In mathematical terms, the matrix representation of (7) leads to the equivalent non-commutative matrix multiplication as the composition. Fig. 3 graphically illustrates the interpretation of left and right composition.

Amongst the list of various representation for $f \in SE(3)$ are the so called minimal representations. A matrix representation (43) is not minimal, neither is the quaternion representation for rotations. When dealing with optimization or estimation problems, minimal representations are highly favorable. Non-linear constraints exist for over-paramatrized representations, they have to be artificially inforced in the optimization procedure and frequently diminish the convergence performance. Indeed there are many coordinate representations $[\mathbf{\Omega}, \mathbf{T}]$, such as various Euler angles, that are minimal. The variety of coordinates comes from the non-commutative ordering of primitive building blocks (generators) of the $SE(3)$ group, see Appendix.

Seemingly, the choice of a particular coordiante system is arbitrary. However, it has been found numerically that gradient based optimization with Euler angles is not stable. This is due to the gimbal lock [], further is has been reported that incremental Rodrigues vector updates are preferable to an additive update [?], also see Appendix F. The choice of proper coordinates strongly affects the convergence performance of gradient based optimization algorithms.

In parameter estimation the focus lies on estimating the uncertainty distribution given noisy measurements. Our experiment shows that the choice of a coordiante system can introduce considerable bias. Interestingly, it was previously noted that in the context of $SE(3)$ pose estimation, an additive parameter noise model is not applicable [?].

In short we want to advocate the view that all coordinate systems are connected via the exponential representation, a representation in which all coordinates commute, hence the issue of the ordering doesn't exist. This coordinate mimicks a euclidian vector space. The advantage is that we can define a parametric uncertainty representation for estimates in those coordinates analogous for additive random variables.

C. Uncertainty in exponential coordinates

A standard approach to probabiliy theory, is to cast the calculus of probabilites into measure theory. The measure $\mu : SE(3) \to \mathbb{R}$ is defined over a set, here a differentiable manifold, and obeys the rules of a sigma algebra. In euclidian spaces, we have an intuitive picture of multivariate probability distributions. If we somehow could connect $SE(3)$ to a euclidean space we would side step plenty mathematical definitions to justify our approach. Fortunately, this connection can be made by means of the exponential map. However, the map is not globally invertible, there exists only locally the function, $\ln : SE(3) \to T_1 SE(3) \cong \mathbb{R}^6$. For the moment we side step the problem of global invertibility, and define a probability measure $p : T_1 SE(3) \to \mathbb{R}$, we also make use of the compositive nature of the group operation $f = f_0 \cdot \Delta f$,

$$\int_{SE(3)} F(f) d\mu(f) \equiv \int_{\mathbb{R}^6} F(f) p_{f_0}(f_0 \cdot \Delta f)\, d\mathbf{t} d\mathbf{\Omega} \tag{10}$$

where $F : SE(3) \to \mathbb{R}^n$ is an arbitrary well-behaved function and $\Delta f = \exp([\mathbf{\Omega}, \mathbf{t}])$.

Adressing the invertibility problem, in practice we assume that the probability density to find a camera pose $f \in SE(3)$ is mono-modal and has its maximum at f_0. Further we assume that the main weight of localization is narrowly confined around this maximum. Despite that the integration domain in (10) is infinite, the main weight only stems from a narrow region around f_0 and a local connection between $SE(3)$ and $\mathbb{R}^6$ is garantueed. Hence, we choose a gaussian distribution in the local tangent space $T_{f_0} SE(3)$ as a plausable parametric approximation to the "real" underlying probability,

$$p_{f_0}(f) \equiv p(\ln f_0^{-1} \cdot f) \sim \mathcal{N}(\mathbf{0}, \Sigma_{ff_r}^{-1}) \tag{11}$$

Put verbally, the probability (11) to find the camera pose $f = f_0 \cdot \Delta f$ is defined by the probability to find a right deviation vector (generator) $\ln \Delta f \in se(3) \cong \mathbb{R}^6$,

$$p(\ln \Delta f) \propto \exp(-\frac{1}{2} [\mathbf{\Omega}, \mathbf{t}]\, \Sigma_{ff_r}^{-1}\, [\mathbf{\Omega}, \mathbf{t}]^T), \tag{12}$$

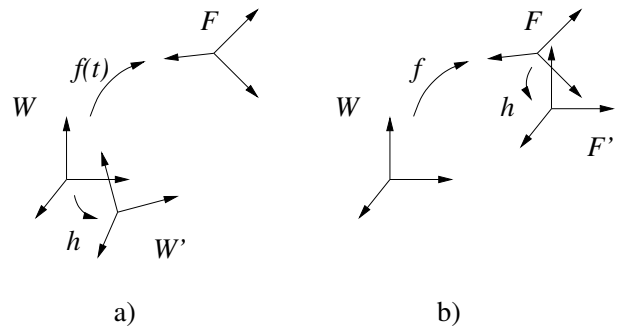

Fig. 3. If $f(t) \in SE(3)$ describes the motion of the rigid body F in time t with respect to frame W, then $h^{-1} \cdot f(t)$ with $h = (\mathbf{R}_h, \mathbf{T}_h)$ is that very same motion seen from frame W'. We could interpret the case a) as a change in the inertial frame $W \to W'$. The effects of a right composition are illustrated in figure b). Suppose F is the space-station, then we could describe the launch $h(t)$ of a rocket F' with respect to frame F. When the rocket dynamics is observed from the world frame W we simply have to calculate $f(t) \cdot h(t)$. Hence a right hand composition effectively mediates a change in the ridig body frame $F \to F'$.

where $\Sigma_{ff_r}^{-1}$ is the real valued 6×6 inverse covariance matrix. We can interpret the vector $\ln \Delta f$ as the geodesic distance on the manifold $SE(3)$ between the points f and f_0.

For notational purpose, we can concisely write an uncertainty frame with mean f_0 and gaussian error model for the deviation Δf as $\hat{f} = (f_0, \Sigma_{ff_r})$. The covariance matrix Σ_{ff_r} has the subscript r to explicitly denote that right-hand composition $f_0 \cdot \Delta f$ is used. Equally well, we could have choosen to compose the "noise" term Δf to the left of f_0. It can be shown, that there is a correponding "left" distribution,

$$f = \Delta f \cdot f_0, \quad \Sigma_{ff_l}^{-1} = \mathsf{J}_{\mathrm{lr}}^T(f_0)\, \Sigma_{ff_r}^{-1}\, \mathsf{J}_{\mathrm{lr}}(f_0), \tag{13}$$

where $\mathsf{J}_{lr}(f_0) \in \mathbb{R}^{6\times 6}$ is the Jacobian matrix associated with the Lie-Derivative at point f_0, see Appendix B. The distinction between "left" and "right" distribution stems from the nature of the non-commutative composition of group elements of a differentiable manifold. For commutative groups both distributions coincide!

D. Transformation of frame and point probability densities

It is clear from the definitions above that a left composition with g (frame transformation) of an uncertainty frame with a right hand noise model $\hat{f} = (f_0, \Sigma_{ff_r})$, will not affect the noise model of the compounded frame, $g \cdot f_0 \cdot \Delta f$. This is in accordance with the composite noise model introduced in [?]. But what happens, if the left composition is also an uncertainty frame? It can be shown that the compostion of two uncertainty frames $\hat{g} = (g_0, \Sigma_{gg_r})$ and $\hat{f} = (f_0, \Sigma_{ff_r})$ yields

$$\begin{aligned} \hat{h} = \hat{g} \circ \hat{f} \;\longrightarrow h_0 &= g_0 \cdot f_0\,, \\ \Sigma_{hh_r} &= \Sigma_{ff_r} + \mathsf{J}_{\mathrm{lr}}(f_0)\Sigma_{gg_r}\mathsf{J}_{\mathrm{lr}}^T(f_0). \end{aligned} \tag{14}$$

Eqn. (14) simpilfies to the abovementioned case if $\hat{g}$ is a certain frame transformation with $\Sigma_{gg_r} = 0$. Similarly, we can connect the uncertainty frame $\hat{f}$ to its inverse $\hat{f}^{-1}$,

$$\hat{f} = (f_0, \Sigma_{ff_r}) \rightarrow \hat{f}^{-1} = (f_0^{-1}, \mathsf{J}_{\mathrm{lr}}^{-1}(f_0)\Sigma_{ff_r}\mathsf{J}_{\mathrm{lr}}^{-T}(f_0)). \tag{15}$$

Another important operation of frames is their action on 3D points (8). Given a gaussian probability distribution for a 3D landmark $p(\mathbf{x})$ we like to know its distribution when we apply a certain frame f_0. It is easily shown that the following holds

$$\begin{aligned} p(\mathbf{x}) \sim \mathcal{N}(\bar{\mathbf{x}}, \Sigma_{xx}) \;\rightarrow\; p(\mathbf{x}_{f_0}) &\sim \mathcal{N}(\bar{\mathbf{x}}_{f_0}, \Sigma_{xx_{f_o}})\,, \\ \Sigma_{xx_{f_o}} &= \mathsf{R}_{f_0}\Sigma_{xx}\mathsf{R}_{f_0}^T. \end{aligned} \tag{16}$$

Finally, we consider the case where a 3D landmark distribution is transformed into an uncertainty frame $\hat{f} = (f_0, \Sigma_{ff_r})$ and we "marginalize" over the frame uncertainty,

$$p(\mathbf{x}_{\hat{f}}) = \int_{\mathbb{R}^6} p(\mathbf{x}_{f_0\cdot\Delta f}) p(\ln \Delta f)\, d\mathbf{t} d\mathbf{\Omega}\,. \tag{17}$$

Despite that the probability distributions in the integrand (17) are gaussian, the resulting marginalized distribution is in general non-gaussion. The integral can be computed in approximation, then the following holds

$$p(\mathbf{x}_{\hat{f}}) \approx \mathcal{N}(\bar{\mathbf{x}}_{f_0}, \Sigma_{xx_{f_o}} + \mathsf{J}_\star \Sigma_{ff_r}\mathsf{J}_\star^T), \;\; \mathsf{J}_\star = \mathsf{R}_{f_0}\mathsf{J}_\times(\mathbf{x}), \tag{18}$$

where we use the defininiton, $\mathsf{J}_\times(\mathbf{x}) = [-[\mathbf{x}]_\times, \mathsf{I}]_{3\times 6}$.

IV. Application to SFM problem

Structure form Motion (SFM) stands for a set of problems to infer the 3 dimensional geometry of an underlying natural scene from a sequence of camera views alone. The camera observations are 2 dimensional in their projective nature. To a high degree it is arguable what constitutes a good observation or measurement. Various heuristic approaches have been taken, most commonly used are 2D point features that are identifyable in various camera views, but also line features or higher polygonal shapes have been proposed.

In this paper, we will not address the data association problem, that is how to decide whether feature observations in various camera views belong to the same underlying geometric object. In the following we deal only with point features and a static scene, meaning that only observer camera motion is allowed. In principle, inference in the Bayesian framework can be extended to any measurement and handle non-static scenes. However, this is beyond the scope of this paper.

In the next section we present the traditional F-matrix approach which is strictly applicable only to 2 camera views and point features. Following that, we address the inference process in the Bayesian framework and conclude with numerical experiments.

A. F-matrix estimation methods

Literature on F-matrix methods and epipolar geometry is certainly too abundant to be cited here, hence we refer to [?], [?] for details and summarize only a few results in the following.

When two camera poses are precisely known, a landmark seen in both views, $\mathbf{p} = [u, v, 1]$ in camera 1 and its corresponding view $\mathbf{q}$ in camera 2, can be readily triangulated. The position of the landmark in 3D space is easily inferred. The consistent triangulation of landmarks in 3 dimensions constitutes a strong constraint for the relative camera pose. Therefore, it is possible to determine the relationship between 2 camera poses from a set of at least 5 landmark measurements. The constraint is effectively stated in a linear relationship, the fundamental matrix[3] $\mathbf{F}$,

$$\mathbf{q}_i^T \mathbf{F} \mathbf{p}_i = 0\,. \tag{19}$$

We note that there is no equivalent constraint in 2 dimensions, the reason is for any camera pose we can find a consistent triangulation.

[3] the linear constraint requires at least 8 point pair correspondences.

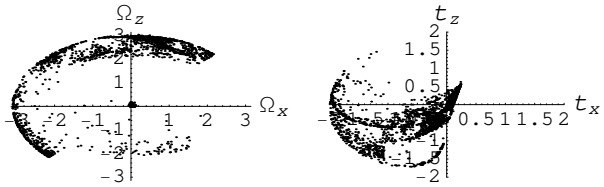

Fig. 4. Sample distribuition (2000 samples) of Least Median Squares estimates for rotation (left) and translation (right) from 68 data points corrupted with gaussian noise with $\sigma = 0.8\%$ of image dimensions

The fundamental matrix $\mathbf{F}$ encodes the relative heading and position of camera 2 with respect to the first one. Hence, we can assume camera 1 at the origin with heading into the positive Z-axis. The distance λ between the two cameras cannot be determined from 2 views only, since any $\mathbf{F}_\lambda = \lambda[\mathbf{T}]_\times \mathbf{R}$ is a valid fundamental matrix with (19). Not only is there a scale ambiguity, also the decomposition into translation $\mathbf{T}$ and rotation $\mathbf{R}$ is four-fold degenerate (if we consider the scale $\lambda > 0$) and has to be resolved by other arguments.

The exact algebraic relationship (19) between 2 views cannot be maintained for noisy measurements. Various estimators have been proposed that extend (19) to noise and outlier prone measurements. Most frequently used are the RANSAC and Least Median of Squares (LMedS) method. Both methods are based on a distance metric $d^2(\mathbf{q}, \mathbf{F}\mathbf{p})$. The LMedS method yields the smallest value for the median of squared residuals, computed for the entire data set

$$M_J = \text{median}\, r_i^2, \quad r_i^2 = d^2(\mathbf{q}_i, \mathbf{F}_J \mathbf{p}_i) + d^2(\mathbf{p}_i, \mathbf{F}_J^T \mathbf{q}_i), \tag{20}$$

where $\mathbf{F}_J$ is based on a small sub-sample J of the entire data set. The estimate $\mathbf{F}_J$ pertaining to the minimum M_J is considered the best robust estimate. Even though LMedS can handle up to 50% outlier with a sufficient number of sub-samples, the algorithm performs poorly on measurements corrupted with gaussian noise.

In [?] it is suggested to amend the result from the LMedS method by an additional M-estimator step. Fig. 4 shows a projected sample distribution where the original zero noise data has been corrupted with gaussian noise with $\sigma = 0.8\%$ of image dimensions. Although the LMedS estimator with addtional reweighted least squares achieves some degree of robustness, tries to concentrate the sample around the zero noise estimate, we still have significant samples "far" away from the undisturbed estimates. For an image with 512 pixels dimensions, 0.8% noise corresponds to ± 2 pixel error. The common problem with the F-matrix estimators remains, even in absence of gross outliers but a larger set of "lesser" quality measurements doesn't yield a sufficiently well behaved estimator distribution. The statistical distribution in Fig. 4 is not sufficiently described by a second momement approximation.

Next, we discuss the estimation process in the Bayesian framework, as "fitting" a model to the observation.

B. Bayesian estimation of joint distribution

The joint probability distribution $p(f, \mathbf{x}|\mathbf{z})$ for camera pose f and 3D model $\mathbf{x}$ is to be inferred from observations $\mathbf{z}$. With Bayes rule we can write in the case of factorizable priors,

$$p(f, \mathbf{x}|\mathbf{z}) \overset{\text{Bayes}}{\propto} p(\mathbf{z}|f, \mathbf{x}) p(\mathbf{x}) p(f). \tag{21}$$

The observation of a single landmark $\mathbf{z}$ causes a mutual dependence of camera pose f and landmark location $\mathbf{x}$. We assume the following observation model for the likelihood

$$p(\mathbf{z}|f, \mathbf{x}) \propto \exp[-\frac{1}{2}(\mathbf{z} - \pi(f \star \mathbf{x}))^T \Sigma_{zz}^{-1} (\mathbf{z} - \pi(f \star \mathbf{x}))], \tag{22}$$

where $\pi(\mathbf{x}') = [x_1'/x_3', x_2'/x_3', 1]$ is the projective function. Further, we may assume that the priors are given as

$$p(\mathbf{x}) \sim \mathcal{N}(\mathbf{x}_0, \Sigma_{xx}), \quad p(f) \sim \mathcal{N}(f_0, \Sigma_{ff_r}). \tag{23}$$

In general the probability distribution (21) can be multimodal and have significant non-gaussian moments. Ensemble representations are adequate to handle the general case. However, for mono-modal cases we promote the view that it is sufficient to consider the MAP estimate of (21). Practically, the MAP estimate is obtained using non-linear optimization techniques based on iterative local linearizations combined with a gradient decent type update step.

1) linearization of observation model:: Linearization of the observation function π in (22) at $\mathbf{x}_t' = f_t \star \mathbf{x}_t$ yields,

$$\Delta \mathbf{z}_t = \mathbf{z} - \pi(f_t \star \mathbf{x}_t), \quad \mathsf{P}_t = \left[\frac{\partial \pi}{\partial \mathbf{x}_t'}\right], \tag{24}$$

particularly we have

$$\pi(\mathbf{x}') = [u, v, 1] \quad \mathsf{P} = \frac{1}{x_3'} \begin{pmatrix} 1 & 0 & -u \\ 0 & 1 & -v \end{pmatrix}, \tag{25}$$

where the singularity at $x_3' = 0$ can be eliminated by a view frustum. The effects of a finite field of view and a shift between optical and geometrical center of the sensor is modeled by a simple calibration matrix K. With the substitution $\mathbf{z} \to \mathsf{K}^{-1}\mathbf{z}$ and $\Sigma_{zz}^{-1} \to \mathsf{K}^T \Sigma_{zz}^{-1} \mathsf{K}$ all the following considerations remain valid.

The derivatives of the transformed points with respect to 3D landmarks $\mathbf{x}$ and a right deviation of the camera pose $f \to f \cdot \delta_r f$ is compactly written as,

$$\frac{\partial (f \star \mathbf{x})}{\partial \mathbf{x}} = \mathsf{R}_f, \quad \frac{\partial (f \star \mathbf{x})}{\partial \delta_r f} = \mathsf{R}_f \, \mathsf{J}_\times(\mathbf{x}), \tag{26}$$

where the defininiton for $\mathsf{J}_\times(\mathbf{x})$ was given in (18).

2) linearization of prior:: The prior probabilitiy distribution (23) has to be linearized also at $[f_t, \mathbf{x}_t]$. Firstly, let's address the 3-D landmark prior distribution $p(\mathbf{x})$,

$$p(\Delta\mathbf{x}_0) \propto \exp(-\frac{1}{2}\Delta\mathbf{x}_0\Sigma_{xx}^{-1}\Delta\mathbf{x}_0), \quad \Delta\mathbf{x}_0 = \mathbf{x} - \mathbf{x}_0. \quad (27)$$

The probability of a deviation $\Delta\mathbf{x}_t$ from the point $\mathbf{x}_t$ is easily derived from the displacement $\Delta\mathbf{x}_0$ seen from the origin $\mathbf{x}_0$,

$$\Delta\mathbf{x}_0 = \mathsf{I}\,\Delta\mathbf{x}_t + (\mathbf{x}_t - \mathbf{x}_0), \quad (28)$$

with I the notation for the identity matrix. Then, the probability $p(\Delta\mathbf{x}_t)$ to find such a displacement $\Delta\mathbf{x}_t$ simply is

$$p(\Delta\mathbf{x}_t) \sim \mathcal{N}(\mathbf{x}_0 - \mathbf{x}_t, \Sigma_{xx}). \quad (29)$$

Eqn. (29) can be interpreted in a particular way: Changing the view point from $\mathbf{x}_0$ to $\mathbf{x}_t$ from which we consider the deviation $\Delta\mathbf{x}_t$, preserves the Mahalanobis metric Σ_{xx}^{-1}. The Mahalanobis metric can be thought as being attached to all points of the $\mathbb{R}^3$ space. A change of the origin in this space will keep the metric unaltered.

Next we address the pose prior $p(f)$ which is defined over the $SE(3)$ manifold. The probability to find f is given by $p(\ln f_0^{-1} \cdot f)$ the probability measure in the euclidian tangent space $T_1SE(3)$ with local Mahalanobis metric $\Sigma_{ff_r}^{-1}$.

Analogously, we consider a small right deviations Δf_t

$$\Delta f_t = (\mathsf{R}(\delta\boldsymbol{\omega}_f), \delta\mathbf{T}_f), \quad \ln\Delta f_t = [\delta\boldsymbol{\omega}_f, \delta\mathbf{T}_f] \to 0, \quad (30)$$

from the new view point f_t. The prior probability to find such a deviation is

$$p(\Delta f_t) = p(\ln g \cdot \Delta f_t), \quad g = f_0^{-1} \cdot f_t = (\mathsf{R}(\boldsymbol{\omega}_g), \mathbf{T}_g). \quad (31)$$

In general, g will not be close to identity!

We note, that $\ln g = [\boldsymbol{\omega}_g, \mathbf{t}_g]$ can be interpreted as the geodesic to reach the point f_t from f_0. Rigorously, $\ln g \in T_1SE(3)$ represents the tangent vector at identity. In the small deviation limit (30) $\ln g \cdot \Delta f_t = [\boldsymbol{\omega}_{\mathrm{id}}, \mathbf{t}_{\mathrm{id}}]$ can be linearized with,

$$\begin{bmatrix} \boldsymbol{\omega}_{\mathrm{id}} \\ \mathbf{t}_{\mathrm{id}} \end{bmatrix} = \underbrace{\begin{pmatrix} \mathsf{J}_{\omega_r}(\boldsymbol{\omega}_g) & \\ \mathsf{J}_t(\boldsymbol{\omega}_g, \mathbf{t}_g) & \mathsf{J}_{\omega_r}(\boldsymbol{\omega}_g) \end{pmatrix}}_{\mathsf{J}_{g_r}} \begin{bmatrix} \delta\boldsymbol{\omega}_f \\ \delta\mathbf{T}_f \end{bmatrix} + \begin{bmatrix} \boldsymbol{\omega}_g \\ \mathbf{t}_g \end{bmatrix}. \quad (32)$$

Eqn. (32) should be seen in direct analogy to (28). For small $[\boldsymbol{\omega}_g, \mathbf{t}_g]$, the Jacobien J_{g_r} will be close to the identity matrix, then Eqn. (32) simplifies to a simple vector translation formula (28). More details about the Wei-Normann Jacobian $[\mathsf{J}_{g_r}]_{6\times6}$ can be found in the Appendix D.

Then, the probability to find a right deviation Δf_t from the camera pose f_t can be shown to be

$$p(\Delta f_t) \sim \mathcal{N}(f_t^{-1} \cdot f_0, {\mathsf{J}_{g_r}}^{-T}\Sigma_{ff_r}{\mathsf{J}_{g_r}}^{-1}). \quad (33)$$

In summary, the intrinsic structure of the $SE(3)$ manifold induces a special vector translation formula (32). A change in the view point $f_0 \to f_t$ effectively mediates a change in the Mahalanobis metric, $\Sigma_{ff_r}^{-1} \to {\mathsf{J}_{g_r}}^T\Sigma_{ff_r}^{-1}\mathsf{J}_{g_r}$. This is in contrast to the constant metric case (29).

3) information matrix update:: The goal is to obtain a MAP estimate to (21). Our approach for finding optimal model parameters in the case of a non-linear observation functions is similar to IEKF methods. However, we mainly use inverse covariance matrices for the update equations. Linearizing the observation function (22) at $f_t \star \mathbf{x}_t$ and the prior at $g = f_0^{-1} \cdot f_t$ with the tangent vector $\mathbf{g} = \ln g$, results in the following central matrix equation to update jointly the camera pose f_t and 3D landmark $\mathbf{x}_t$,

$$\left(\Sigma_{\mathrm{prior}}^{-1} + \Sigma_{\mathrm{msr}}^{-1}\right)\begin{bmatrix} \Delta\mathbf{f}_t \\ \Delta\mathbf{x}_t \end{bmatrix} = \mathsf{J}_{\mathrm{obs}}\Delta\mathbf{z}_t - \Sigma_{\mathrm{prior}}^{-1}\begin{bmatrix} {\mathsf{J}_{g_r}}^{-1}\mathbf{g} \\ \mathbf{x}_t - \mathbf{x}_0 \end{bmatrix} \quad (34)$$

where we use the following notation for the prior and measurement information matrix

$$\begin{aligned} \Sigma_{\mathrm{prior}}^{-1} &= \begin{pmatrix} {\mathsf{J}_{g_r}}^T\Sigma_{ff_r}^{-1}\mathsf{J}_{g_r} & \\ & \Sigma_{xx}^{-1} \end{pmatrix}, \\ \Sigma_{\mathrm{msr}}^{-1} &= \begin{pmatrix} \mathsf{JMJ}(f_t, \mathbf{x}_t) & \mathsf{JM}(f_t, \mathbf{x}_t) \\ \mathsf{MJ}(f_t, \mathbf{x}_t) & \mathsf{M}(f_t, \mathbf{x}_t) \end{pmatrix}. \end{aligned} \quad (35)$$

Definitions for $\mathsf{J}_{\mathrm{obs}}$ and the entries of the measurement information matrix $\Sigma_{\mathrm{msr}}^{-1}$ are given in the Appendix G. The camera pose and 3D landmark estimates are updated with,

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \Delta\mathbf{x}_t, \quad f_{t+1} = f_t \cdot \exp(\Delta\mathbf{f}_t). \quad (36)$$

With the new estimates we resume linearization at $[f_{t+1}, \mathbf{x}_{t+1}]$ then solving (34) till sufficient convergence is reached. Eqns. (34,36) represent the implementation of the Gauss-Newton iterateration for the joint estimates of camera pose and 3D landmark with priors.

In general, the measurement matrix $\Sigma_{\mathrm{msr}}^{-1}$ is rank deficienct. Then, solving for (34) has to done carefully. Frequently, the pseudo inverse [6] is used to avoid problems with soft modes. Also trust-region algorithms such as traditional Levenberg-Marquardt or gauge-invariant version thereof [7] are applied to improve the convergence rate. Those modes are a manifestation of the degeneracies (ambiguities) in the inference process. The use of a prior can eliminate unwanted ambiguities.

C. Multiple cameras and landmarks (bundle adjustment)

The update equation (34) generalizes to the case with multiple camera views and a set of 3D landmarks, $\{f^i, \mathbf{x}^j\}$. We denote the connection matrix $\mathsf{c}^{i,j}$,

$$\mathsf{c}^{i,j} = \begin{cases} 1, & \text{if landmark } j \text{ is seen in camera } i \\ 0, & \text{otherwise.} \end{cases} \quad (37)$$

Bundle adjustment solves for a set of m camera poses and n 3D landmark position simultaneously. It consists of (non-linear) minimization of the likelihood function

$$-\ln L(\mathbf{z}^{i,j}|\{f^i, \mathbf{x}^j\}) \;\propto\; \sum_{i=1}^{m}\sum_{j=1}^{n} \mathsf{c}^{i,j}\,|\underbrace{\mathbf{z}^{i,j} - \pi(f^i \star \mathbf{x}^j)}_{\Delta\mathbf{z}^{i,j}}|^2 . \quad (38)$$

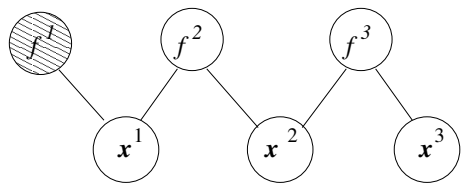

Fig. 5. The moralized graph for a bundle adjustment with 2 free cameras and 3 landmarks. Camera f^1 fixes the world-frame and is not updated in the minimization step.

Initialization $[\mathbf{f}_0^i, \mathbf{x}_0^j]$ is a critical issue to ensure convergence towards the global minimum. It is mostly done from the results obtained by F-matrix methods. Generally, the measurement matrix is highly structured and sparse. As an example, the matrix to Fig. 5 can be found in the appendix. The sparseness can be exploited to find the solution to (38) computationally efficient, [8]. However, the use of a first order approximation (34) limits greatly the convergence radius, since the observation function (38) is essentially non-convex.

1) Ambiguities: The likelihood function (38) can be shown to have various continouus softmodes. The obvious ones are the anchor and scale softmode. The former one arises because of the interdependence between observer and observation. Without prior information, there is no predefined orientation nor origin. We can arbitrarily choose the first camera as an anchor to set the reference frame. The scale softmode is solely due to the projective nature of the observation function π. We cannot infer the the absolute distance scale between 3D landmarks and camera from observations alone, therefore the following transformation leaves the likelihood function (38) unalterd

$$\mathbf{x} \to \mathbf{x}' = s\,\mathbf{x}, \quad f = (\mathbf{R}_f, \mathbf{T}_f) \to f' = (\mathbf{R}_f, s\,\mathbf{T}_f), \quad (39)$$

for $s > 0$ the scale factor.

2) Hardcoded constraints, Master-Slave cameras: Frequently, a stereo camera rig is used in robotic applications. The relative camera pose Δ_{ms} between the stereo cameras is well known and independent inference of their poses should not be attempted. The Bayesian inference process is easily adaptable to this case using a master and slave camera concept. We simply make the slave camera pose algebraically dependent on a master camera:

$$f_{\text{slave}} = \Delta_{\text{ms}} \cdot f_{\text{master}}, \quad \frac{\partial f_{\text{slave}} \star \mathbf{x}}{\partial \delta_r f} = \mathsf{R}_{f_{\text{slave}}}\, \mathsf{J}_{\times}(\mathbf{x}), \quad (40)$$

a right deviation in the master camera $f_{\text{master}} \to f_{\text{master}} \cdot \delta_r f$, leads to a dependent change in the transformation into the slave camera.

The scale softmode (39) can be eliminated with the use of a well calibrated stereo camera rig. The distance between both cameras serves as a yardstick to fix the distance scale s.

D. The marginalized MAP estimator, marginalization over 3D environment

As discussed in the previous chapter, the joint MAP estimate of n camera poses and m 3D landmarks leads to a algebraically coupled system with the measurement matrix Σ_{msr}^{-1} of dimensions $(6n+3m)^2$. Successive observations can lead to a large number of landmarks, therefore the storage requirements and the numerical inversion of a large matrix become computationally unpractical.

However, the product rule

$$p(f, \{\mathbf{x}\}|\mathbf{z}) = p(f|\mathbf{z}) \prod_j p(\mathbf{x}^j|f, \mathbf{z}) \quad (41)$$

facilitates an easier way to find the MAP estimate to (41) using a two step scheme Step 1, estimate the optimal 3D landmarks $\mathbf{x}_{\text{opt}}$ given the observation and camera pose, then we use the saddle point method to marginalize over the 3D environment

$$p(f|\mathbf{z}) = \prod_j \int p(f, \mathbf{x}_{\text{opt}}^j|\mathbf{z})\, d\mathbf{x}_{\text{opt}}^j. \quad (42)$$

Step 2, we apply the Gauss-Newton method to find the optimal camera pose, thus reducing the dimensions of the algebraic system (34) to $6n$ dimensions. Subsequent iteration of the two-step scheme yields the desired MAP estimate.

This two-step scheme is computational efficient, finding the optimal landmarks $\mathbf{x}_{\text{opt}}$ is linear in m, as well is the marginalization step.

V. Experimental Results

We have tested the mariginalized MAP algorithm for various real natural environment scenarios and compared it against the bundle adjustment technique. Fig. 6 shows an example for two matched images. We found that convergence performance of the bundle adjustment procedure (joint estimate of pose and 3D landmarks) versus the marginalized MAP algorithm is roughly similar in cases where the initialization for bundle-adjustment procedure is within the basin of attraction, see Fig. 6. We note that generally 10 iteration steps are sufficient for convergence. However, the region of convergence is considerable larger for the marginalized MAP algorithm since the non-convexity of the cost-function in the likelihood term is adequately dealt with. Secondly, the joined estimate of camera pose and 3D landmarks requires operations on large matrices (here 68 landmarks and 6 DOF for camera requires a 210×210 matrix), whereas the largest matrix in the marginalized MAP algorithm is 6×6.

Gaussian noise is imposed on the feature point observations in order to study the camera pose estimate distribution. Fig. 8 shows the discrete point sample distribution for the camera pose estimates obtained by the marginalized MAP algorithm. This sample distribution should be seen in direct contrast to the estimate distribution obtained by F-matrix methods, see Fig. 4.

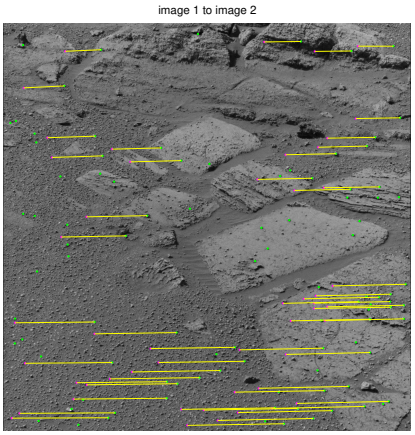

Fig. 6. Experiments were conducted for typical natural terrain scenarios. Firstly, feature points are defined using image operations. Matching of features in a sequence of images is achieved in a second step. Here, matches are represented as yellow lines. Mismatches will produce gross outliers, as well as smaller variations in feature point localization impose some measurement noise.

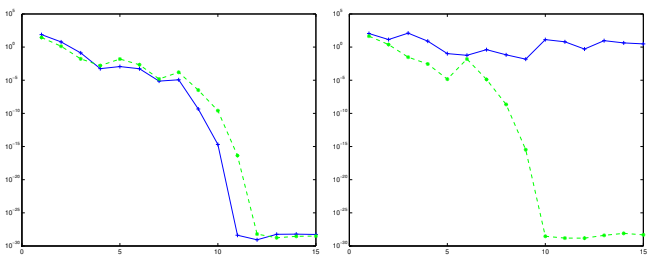

Fig. 7. Performance of bundle-adjustment versus marginalized MAP estimator. The logarithmic likelihood is plotted on a log scale versus iteration steps. Bundle-adjustment (blue solid lines) performs slightly better than the marginalized MAP algorithm in the case when bundle-adjustment converges, see left hand figure. However, the convergence radius for bundle-adjustment is rather small due to the non-convexity of the cost function. Here, the marginalized MAP overcomes this problem and converges rapidly, see right hand figure.

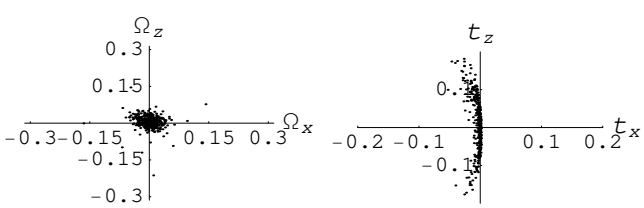

Fig. 8. Discrete point sample distribution of camera pose estimates in exponential coordinates for rotation (left) and translation (right) from matched data points, seen in Fig. 6 corrupted with gaussian noise with $\sigma = 0.8\%$ of image dimensions. This depiction should be seen in contrast to Fig. 4

VI. Conclusion

We have presented a theoretical and numerically implementable framework for uncertainty distributions over the angle and translational space based on the notion of the $SE(3)$ Lie group. Our numerical experiments hint that the probability distributions in exponential coordinates have significant gaussian characteristics which makes it possible to use KF techniques for parameter estimation.

Further, on the application side we have introduced a novel 2 step estimation technique, the marginalized MAP estimator, which is numerically more stable and computationally much more efficient than bundle-adjustment, the traditional technique for camera pose estimation from images. Future work will focus on a recursive estimation scheme as well as proper outlier-detection based on MAP-SAC.

Appendix

A group element $g \in SE(3)$ can be represented by a real valued 4×4 matrix

$$g \cong (\mathsf{R}(\mathbf{\Omega}_g), \mathbf{T}_g) \equiv \begin{pmatrix} \mathsf{R}(\mathbf{\Omega}_g) & \mathbf{T}_g \\ 0\,0\,0 & 1 \end{pmatrix} \tag{43}$$

and group operation is endowed as usual by matrix multiplication. Throughout this appendix, $[\mathbf{v}]_\times$ denotes the skew-symmetric matrix associated with a vector $\mathbf{v} \in \mathbb{R}^3$, defined by $[\mathbf{v}]_\times \mathbf{w} = \mathbf{v} \times \mathbf{w}$ for all $\mathbf{w} \in \mathbb{R}^3$.

A. Exponential and logarithmic map

Any matrix of the structure (43) can be obtained by the exponential map

$$\exp : \mathbb{R}^6 \to SE(3),\ [\mathbf{\Omega}_g, \mathbf{t}_g]_{L_1} \to \exp(\hat{\mathbf{H}} \cdot \mathbf{\Omega}_g + \hat{\mathbf{T}} \cdot \mathbf{t}_g) = g, \tag{44}$$

where the coordinates $[\mathbf{\Omega}_g, \mathbf{t}_g]_{L_1} \in \mathbb{R}^6$ are called the Lie-Cartan coordinates of the first kind. We usually will omit the notation subscript L_1 when dealing with those exponential coordinates. The exponential map is globally well defined, and can be depicted geometrically as the path connecting the identity element $1 \in SE(3)$ with g. In (44), the basis elements of the 6-dimensional vector space $\mathbb{R}^6 \cong T_1 SE(3)$ are compactly denoted as,

$$\hat{\mathbf{H}} \cdot \boldsymbol{\omega} = \mathsf{H}_x \Omega_x + \mathsf{H}_y \Omega_y + \mathsf{H}_z \Omega_z, \quad \hat{\mathbf{T}} \cdot \mathbf{t} = \mathsf{T}_x t_x + \mathsf{T}_y t_y + \mathsf{T}_z t_z, \tag{45}$$

where each element of $\hat{\mathbf{H}}$ and $\hat{\mathbf{T}}$ are real 4×4 matrices, also called the generators of the $SE(3)$ group. In conjunction with the Lie-product (commutator), $[\mathsf{A}, \mathsf{B}]_c = \mathsf{AB} - \mathsf{BA}$ [4], the vector space becomes the $se(3)$ Lie algebra,

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk} \mathsf{H}_k,\ [\mathsf{H}_i, \mathsf{T}_j]_c = \varepsilon_{ijk} \mathsf{T}_k,\ [\mathsf{T}_i, \mathsf{T}_j]_c = 0. \tag{46}$$

The commutator can be seen as the adjoint operator,

$$\mathrm{ad}(\mathsf{A})\mathsf{B} = [\mathsf{A}, \mathsf{B}]_c, \quad \mathrm{ad}(\mathsf{A})^k(\mathsf{B}) = [\mathsf{A}, \ldots, [\mathsf{A}, [\mathsf{A}, \mathsf{B}]_c]_c, \tag{47}$$

[4] the square bracket notation is frequently used without the subscript c, we use it here because we already defined $[\boldsymbol{\omega}, \mathbf{v}]$ as the concatenation of 2 vectors.

which gives a powerful notation to represent k interlocked commutators as shown in (47).

Frequently, we will have to deal with a factorized representation of (44)

$$(\mathsf{R}_z(\phi)\mathsf{R}_y(\vartheta)\mathsf{R}_z(\psi), \mathbf{T}) = \exp(\hat{\mathbf{T}}\cdot\mathbf{T}) \\ \exp(\mathsf{H}_z\phi)\exp(\mathsf{H}_y\vartheta)\exp(\mathsf{H}_z\psi), \quad (48)$$

where $(\mathbf{T}, \phi, \vartheta, \psi)_{L_2}$ are called canonical coordinates of second kind [5]. Due to the non-trivial algebra structure (46) the connection between first and second kind coordinates is not a simple one. However, we can give a simple commutation rule for the following case

$$\exp(\hat{\mathbf{T}}\cdot\mathbf{T})\exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega}) = \exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega})\exp(\hat{\mathbf{T}}\cdot\mathsf{R}^{-1}(\mathbf{\Omega})\mathbf{T}), \quad (49)$$

which can be derived from the general conjugation rule

$$g^{-1}\,\exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega}+\hat{\mathbf{T}}\cdot\mathbf{T})\,g = \exp(g^{-1}\,(\hat{\mathbf{H}}\cdot\mathbf{\Omega}+\hat{\mathbf{T}}\cdot\mathbf{T})\,g) \quad (50)$$

where g in the exponent of (50) is interpreted as a 4×4 matrix, according to (43).

In particular, for the exponential map (44) we can give the following Eqns.

$$\begin{aligned} \mathsf{R}(\mathbf{\Omega}) &= \mathsf{I} + \sin\Theta\mathsf{H} + (1-\cos\Theta)\mathsf{H}^2, \quad (51)\\ \mathbf{T} &= \mathsf{S}(\mathbf{\Omega})\,\mathbf{t}, \quad \mathsf{S}(\mathbf{\Omega}) = \mathsf{I} + \mathsf{H}^2 + \frac{1}{\Theta}(\mathsf{I}-\mathsf{R})\mathsf{H}, \end{aligned}$$

where we use the notation

$$\Theta = |\mathbf{\Omega}|,\ \mathbf{r} = \mathbf{\Omega}/\Theta, \quad \mathsf{H} = \hat{\mathbf{H}}\cdot\mathbf{r}. \quad (52)$$

As seen from the Rodrigues formula (51), $\mathsf{R}(\mathbf{\Omega})$ is 2π-periodic in Θ. This implies that a change in the normal vector direction can be compensated by a 2π offset, $(\mathbf{r},\Theta) \cong (-\mathbf{r}, 2\pi-\Theta)$. Hence, $\mathbf{\Omega}$ uniquely specifies a rotation only when it is restricted within a sphere of radius π. Penetrating the skin of this sphere causes phase reflection, where opposite points on the boundary are identified, $(\mathbf{r},\pi) \equiv (-\mathbf{r},\pi)$. This phenomenon has a precise topological origin in the double-cover structure $SU(2)/\mathbb{Z}_2 \cong SO(3)$, which we develop in the next subsection.

The inverse function to the exponential map is the logarithmic map

$$\ln : SE(3) \to \mathbb{R}^6,\ (\mathsf{R}(\mathbf{\Omega}), \mathbf{T}) \to [\mathbf{\Omega}, \mathbf{t}]_{L_1}, \quad (53)$$

which is well defined for $\Theta < \pi$. As discussed above, the logarithmic map will not be unique at the phase reflection boundary, $\Theta = \pi$. The existence of such boundary will have implications for the topics considered in the following sections.

[5] the ordering of the exponential factors is important, hence there are multiple sets of coordinates, i.e. Euler ZYZ, XYZ...

B. The $SU(2)/\mathbb{Z}_2 \cong SO(3)$ connection

The $SO(3)$ generating algebra H_i and the $SU(2)$ one with generators u_i have very similar structures, meaning there is an isomorphism between the algebras, $\mathfrak{so}(3) \cong \mathfrak{su}(2)$,

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk}\mathsf{H}_k,\ [\mathsf{u}_i, \mathsf{u}_j]_c = 2\varepsilon_{ijk}\mathsf{u}_k\ . \quad (54)$$

Hence, after the rescaling $\tau_i = \frac{1}{2}\mathsf{u}_i$, we obtain $[\tau_i, \tau_j]_c = \varepsilon_{ijk}\tau_k$, so that the correspondence $\mathsf{H}_i \leftrightarrow \tau_i$ defines the Lie-algebra isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

Both H_0 and u_0 are the identity matrices in each of their cases; they are the identity elements of the group representations, not generators of the Lie algebra. The standard matrix representations are:

$$\mathsf{u}_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},\ \mathsf{u}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},\ \mathsf{u}_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad (55)$$

and for the Lie algebra $\mathfrak{so}(3)$ generators of the $SO(3)$ group

$$\mathsf{H}_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathsf{H}_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \\ \mathsf{H}_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Other representations of $\mathfrak{su}(2)$ are possible and standard in Physics, with the algebra structure (54) changing slightly to $[\sigma_i, \sigma_j]_c = 2i\varepsilon_{ijk}\sigma_k$, which introduces the Pauli matrices:

$$\sigma_1 = -i\mathsf{u}_1\,, \quad \sigma_2 = i\mathsf{u}_2\,, \quad \sigma_3 = -i\mathsf{u}_3\,. \quad (56)$$

Regardless of the isomorphic $\mathfrak{su}(2)$ representations, the generated group $SU(2)$ is not isomorphic to $SO(3)$: $SU(2)$ is the simply connected double cover of $SO(3)$. To see this explicitly through the Rodrigues formulas, let

$$\mathsf{u} = \hat{\mathsf{u}}\cdot\mathbf{r} = r_i\,\mathsf{u}_i\,, \qquad \Theta = |\mathbf{\Omega}|\,, \qquad \mathbf{r} = \mathbf{\Omega}/\Theta\,. \quad (57)$$

Since $\mathsf{u}^2 = -\mathsf{u}_0$ for any unit vector $\mathbf{r}$, the matrix exponential in $SU(2)$ truncates to

$$U(\mathbf{\Omega}) = \exp\Big(\tfrac{1}{2}\,\hat{\mathsf{u}}\cdot\mathbf{\Omega}\Big) = \mathsf{u}_0\cos\frac{\Theta}{2} + \mathsf{u}\,\sin\frac{\Theta}{2}\,. \quad (58)$$

The adjoint action of U on the pure-imaginary quaternion subspace induces a rotation $\mathsf{R}(\mathbf{\Omega}) \in SO(3)$:

$$U(\mathbf{\Omega})\,(\hat{\mathsf{u}}\cdot\mathbf{x})\,U(\mathbf{\Omega})^{-1} = \hat{\mathsf{u}}\cdot\big(\mathsf{R}(\mathbf{\Omega})\,\mathbf{x}\big)\,, \\ \mathsf{R}(\mathbf{\Omega})\,\mathbf{x} = \cos\Theta\,\mathbf{x} + (1-\cos\Theta)(\mathbf{r}\cdot\mathbf{x})\,\mathbf{r} + \sin\Theta\,\mathbf{r}\times\mathbf{x}\,. \quad (59)$$

In matrix form this is precisely the Rodrigues formula (51).

Equivalently, writing U in quaternion parametrisation,

$$U = q_0\,\mathsf{u}_0 + q_i\,\mathsf{u}_i\,, \qquad q_0 = \cos\frac{\Theta}{2}\,, \qquad \mathbf{q} = \sin\frac{\Theta}{2}\,\mathbf{r}\,, \qquad q_0^2 + \mathbf{q}^\top\mathbf{q} = 1\,, \quad (60)$$

the induced rotation matrix becomes

$$\mathsf{R}(U) = (q_0^2 - \mathbf{q}^\top\mathbf{q})\,\mathsf{H}_0 + 2\,\mathbf{q}\,\mathbf{q}^\top + 2\,q_0\,[\mathbf{q}]_\times\,. \quad (61)$$

This exhibits $SU(2) \cong S^3$ directly.

a) The covering map.: The surjective homomorphism $\pi\colon SU(2) \to SO(3)$ is therefore given by

$$\pi(U) = \mathsf{R}(U)\,, \qquad \ker\pi = \{\pm \mathsf{u}_0\}\,. \tag{62}$$

Hence

$$SO(3) \cong SU(2)/\{\pm\mathsf{u}_0\} \cong S^3/\mathbb{Z}_2\,. \tag{63}$$

b) The two-to-one identification.: In the exponential parametrisation the double cover appears as

$$U(\mathbf{\Omega} + 2\pi\mathbf{r}) = -\,U(\mathbf{\Omega})\,, \qquad \pi(U) = \pi(-U)\,. \tag{64}$$

Antipodal points in $SU(2) \cong S^3$ therefore represent the same rotation in $SO(3)$. This is the precise topological origin of the phase-reflection phenomenon encountered in the previous subsection: the identification $(\mathbf{r}, \pi) \equiv (-\mathbf{r}, \pi)$ on the boundary of the Rodrigues parameter sphere is the quotient $S^3/\mathbb{Z}_2$ restricted to the equator. Equivalently, $\pi_1(SO(3)) = \mathbb{Z}_2$, whereas $SU(2) \cong S^3$ is simply connected.

c) Inverse map.: Given $\mathsf{R} \in SO(3)$, the rotation angle follows from $\operatorname{Tr}\mathsf{R} = 1 + 2\cos\Theta$ and the axis from the antisymmetric part, $n_k = -\varepsilon_{ijk}\,R_{ij}/(2\sin\Theta)$, after which (58) returns the $SU(2)$ element up to an overall sign. Alternatively, reading off from (61) directly: $q_0 = \pm\frac{1}{2}\sqrt{1+\operatorname{Tr}\mathsf{R}}$ and $q_k = -\varepsilon_{ijk}\,R_{ij}/(4q_0)$ for $q_0 \neq 0$. The sign ambiguity is precisely the $\mathbb{Z}_2$ kernel: no continuous global section $SO(3) \to SU(2)$ exists.

C. Baker-Campbell-Hausdorff formula: composing two finite rotations

Frequently, we obtain a factorized rotation representation,

$$\exp(\Theta_c\mathsf{H}_c) = \exp(\Theta_a\mathsf{H}_a)\exp(\Theta_b\mathsf{H}_b), \tag{65}$$

presented with notation (52). An exponential series expansion of the right hand side of (65) is ordered with H_a to the left and H_b to the right. It is clear that a simple sum $\mathsf{H}_c \neq \mathsf{H}_a + \mathsf{H}_b$ cannot hold because its exponential series expansion is unordered. The Baker-Campbell-Hausdorff (BCH) formula[6] is useful to derive an additive relationship,

$$\begin{aligned}\Theta_c\mathsf{H}_c &= \Theta_b\mathsf{H}_b + \int_0^{\Theta_a} h\left(e^{s\,\mathrm{ad}(\mathsf{H}_a)}e^{\Theta_b\,\mathrm{ad}(\mathsf{H}_b)}\right)\mathsf{H}_a\,ds,\\ &= \Theta_a\mathsf{H}_a + \Theta_b\mathsf{H}_b + \frac{1}{2}\Theta_a\Theta_b[\mathsf{H}_a,\mathsf{H}_b] + \ldots\,.\end{aligned} \tag{66}$$

However, (66) only hands an infinite series expansion. It is quite useful to have a closed expression for the Rodrigues vector $\mathbf{\Omega}_c$ of the resulting rotation $\mathsf{R}_c = \exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega}_c)$. We could compactly denote this operation as $\mathbf{\Omega}_c = \ln\mathsf{R}_a\mathsf{R}_b$.

[6] the BCH formula can be derived using the derivative of exponentials, the resulting integral with function $h(x) = \ln(x)/(x-1)$ is understood as a power series expansion in x.

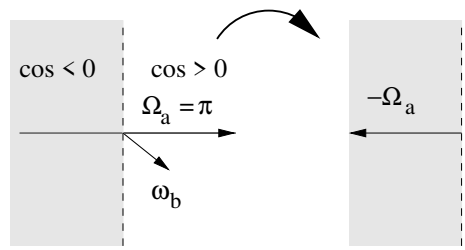

Fig. 9. Phase reflection occurs for $|\mathbf{\Omega}_a| = \pi$. The result of adding another infinitesimal vector $\boldsymbol{\omega}_b$ depends whether it has the same direction $\cos\vartheta > 0$, which will give a resulting vector in the direction $-\mathbf{\Omega}_a$. If $\cos\vartheta < 0$, the resulting vector will be unchanged.

Using the quaternion representation of $SU(2)$ developed in §B, one can show that the following Eqn. holds,

$$\mathbf{\Omega}_c = \frac{\Theta_c}{C}\left(a\,\mathbf{r}_a + b\,\mathbf{r}_b + ab\,\mathbf{r}_a\times\mathbf{r}_b\right), \qquad a = \tan\frac{\Theta_a}{2},\; b = \tan\frac{\Theta_b}{2}. \tag{67}$$

The normalizing factor, $C = \sqrt{A - \varepsilon^2}$, can be given as

$$A = 1 + a^2 + b^2 + a^2b^2, \;\; \varepsilon = (1 - ab\cos\vartheta), \quad \cos\vartheta = \mathbf{r}_a\cdot\mathbf{r}_b\,, \tag{68}$$

and the norm Θ_c

$$\tan\frac{\Theta_c}{2} = C/\,\varepsilon\,. \tag{69}$$

For the trivial case, $\cos\vartheta = \pm 1$ in (68), we simply obtain $\mathbf{\Omega}_c = \mathbf{\Omega}_a + \mathbf{\Omega}_b$. To evaluate the prefactor Θ_c/C in (67) or the primed versions below, we have to pay attention to the following two limit cases:

1) limit $\varepsilon \to 0$, (this implies $A \geq 1$)

$$\frac{\Theta_c}{C} = \frac{\pi}{\sqrt{A}}\,[1 - \frac{2}{\pi}\,\frac{\varepsilon}{\sqrt{A}} + \frac{1}{2}\left(\frac{\varepsilon}{\sqrt{A}}\right)^2 + \mathcal{O}((\varepsilon/\sqrt{A})^3)\,] \tag{70}$$

2) limit $C \to 0$, (this implies $\varepsilon^2 \geq 1$)

$$\frac{\Theta_c}{C} = \frac{2}{\varepsilon}\,[1 - \frac{1}{3}\left(\frac{C}{\varepsilon}\right)^2 + \mathcal{O}((C/\varepsilon)^4)\,]\,. \tag{71}$$

1) Phase reflection at $\Theta = \pi$: Since the logarithmic map loses its uniqueness at $\Theta = \pi$ (cf. the double-cover discussion in §B), we have to be careful to define the vector addition (67). Firstly, we consider $\Theta_a \to \pi-$ which implies $a \to \infty$ and b finite. Secondly a similar statement accounts for the limit $a \to \infty$ and $b \to \infty$. We notice that those two limits, (i) and (ii), can be treated consistently. In both cases a cutline at $\cos\vartheta = 0$ has to be made to account for the phase reflection boundary. The limit process $\Theta_a \to \pi-$ can be depicted graphically, see Fig. 9. Hence, all quantities are properly defined in the limits. Therefore, Eqn. (67) can be used to add finite Rodrigues vectors without employing intermediate rotation matrices or quaternions!

2) Adding two infinitesimal rotations: kinematics of rigid bodies: The rate of change, angular velocity, of $\mathbf{\Omega}$ for an infinitesimal small quantity $dt \to 0$ is $d\mathbf{\Omega} = dt\,\boldsymbol{\omega}$. We can apply the finite addition formula (67) or directly apply the BCH expansion (66) to obtain,

$$d\mathbf{\Omega}_c = dt\,\boldsymbol{\omega}_c = dt\,\boldsymbol{\omega}_a + dt\,\boldsymbol{\omega}_b + \frac{1}{2}dt^2\,\boldsymbol{\omega}_a \times \boldsymbol{\omega}_b\,. \tag{72}$$

As such we recover in (72) the well known fact that angular velocities $\boldsymbol{\omega}$ constitute an algebraic vector space with the normal commutative addition. More exactly, the vector space is known as $T_1SO(3)$.

D. Wei-Norman formula: composing a finite with an infinitesimal rotation

We now derive the right Jacobian J_{ω_r} that relates infinitesimal perturbations in Lie-algebra coordinates to the corresponding group-level variations. We present two routes: first via the Wei-Norman formula using the derivative of exponentials, then a direct computation from the finite addition formula (67).

Again, we consider the rotation product expression in (65). This time we like to describe the infinitesimal environment at $\mathbf{\Omega}_c = \mathbf{\Omega}_a + \Delta\boldsymbol{\omega}$ for a finite left rotation $\mathbf{\Omega}_a$ in dependence of an infinitesimal right rotation $\mathbf{\Omega}_b \to \mathbf{0}$. Then we have to analyze the following directional derivative

$$\frac{d}{ds}\,\exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega}_a + s\,\hat{\mathbf{H}}\cdot\Delta\boldsymbol{\omega}) \;=\; \exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega}_a)\hat{\mathbf{H}}\cdot\mathsf{J}_{\omega_r}^{-1}(\mathbf{\Omega}_a)\Delta\boldsymbol{\omega}, \tag{73}$$

where the right composition Jacobian $\mathsf{J}_{\omega_r}(\mathbf{\Omega}_a) \in \mathbb{R}^{3\times 3}$: $T_{\mathsf{R}(\mathbf{\Omega}_a)}SO(3) \to T_1SO(3)$ describes the mapping from the tangent space at $\mathsf{R}(\mathbf{\Omega}_a)$ to the one at identity. In order to evaluate the left hand side of (73), it is advantageous to consider a "matrix" equation for $g \in SE(3)$ and general $\mathsf{A}, \mathsf{B} \in se(3)$

$$\frac{\partial}{\partial t}g(t,s) = (\mathsf{A} + s\mathsf{B})g(s,t) \;\to\; g(s,t) = e^{t(\mathsf{A}+s\mathsf{B})}. \tag{74}$$

With the help of the Dyson ordering operator $\mathcal{T}$ and the adjoint notation (47), it can be shown that the following holds

$$g(s,t) = e^{t\mathsf{A}}\,\mathcal{T}\exp\left(s\int_0^t e^{-u\,\mathrm{ad}(\mathsf{A})}\,du\;\mathsf{B}\right)\,. \tag{75}$$

Differentiating (75) with respect to s, setting $t = 1$, then applying it to (73) yields

$$u(\Theta_a\,\mathrm{ad}(\mathsf{H}_a))\hat{\mathbf{H}}\cdot\Delta\boldsymbol{\omega} = \hat{\mathbf{H}}\cdot\mathsf{J}_{\omega_r}^{-1}(\mathbf{\Omega}_a)\Delta\boldsymbol{\omega}. \quad u(x) = \int_0^1 e^{-sx}\,ds \tag{76}$$

a relationship to obtain $\mathsf{J}_{\omega_r}(\mathbf{\Omega}_a)$. In fact (76) constitutes a Wei-Norman formula [16], which relates the differential changes of a product of exponentials (65) to the changes of one resulting exponential. However, it is not easy to obtain a closed form expression from (76).

a) Direct route via finite addition.: It shall be noticed that with Eqn. (67) we directly obtain the right Rodrigues Jacobian $\mathsf{J}_{\omega_r}(\mathbf{\Omega}_a)$ by,

$$\mathsf{J}_{\omega_r}(\mathbf{\Omega}_a) = \left.\frac{\partial\mathbf{\Omega}_c}{\partial\mathbf{\Omega}_b}\right|_{\Theta_b=0} = \left.\frac{\partial\mathbf{\Omega}_c}{\partial\Theta_b}\right|_{\Theta_b=0}\mathbf{r}_b + \left.\frac{\partial\mathbf{\Omega}_c}{\partial\mathbf{r}_b}\frac{\partial\mathbf{r}_b}{\partial\mathbf{\Omega}_b}\right|_{\Theta_b=0}. \tag{77}$$

Further, Eqn. (77) can be evaluated as,

$$\mathsf{J}_{\omega_r}(\mathbf{\Omega}_a) = (1-x)\,\mathsf{I} + x\,\mathbf{r}_a\mathbf{r}_a^T + \frac{1}{2}[\mathbf{\Omega}_a]_\times,\; 1-x = \frac{\Theta_a}{2}\tan^{-1}\frac{\Theta_a}{2}. \tag{78}$$

Similarly, we could leave $\mathbf{\Omega}_b$ in the product expansion (65) at a finite value and consider infinitesimal left variations of $\mathbf{\Omega}_a \to \mathbf{0}$. This naturally leads to the complementary left Rodrigues Jacobian

$$\mathsf{J}_{\omega_l}(\mathbf{\Omega}_b) = (1-y)\,\mathsf{I} + y\,\mathbf{r}_b\mathbf{r}_b^T - \frac{1}{2}[\mathbf{\Omega}_b]_\times,\; 1-y = \frac{\Theta_b}{2}\tan^{-1}\frac{\Theta_b}{2}. \tag{79}$$

It shall be emphasised that the Jacobians above don't have singularities and are invertible, since all quantities are well defined for $\Theta \le \pi$ and $\det(\mathsf{J}_{\omega_r}) = (1-x)^2 + \Theta_a^2/4$.

E. Composition Jacobians over $SE(3)$

In close analogy to the previous sections, a composition in $SE(3)$ can be analyzed in the additive framework,

$$(\mathsf{R}(\mathbf{\Omega}_c), \mathbf{T}_c) = (\mathsf{R}_a, \mathbf{T}_a)\cdot(\mathsf{R}_b, \mathbf{T}_b)\,. \tag{80}$$

1) Composing finite motions: With definition of this group composition (9), we can formally write

$$\ln(\mathsf{R}(\mathbf{\Omega}_c), \mathbf{T}_c) = [\ln\mathsf{R}_a\mathsf{R}_b,\; \mathsf{S}^{-1}(\mathbf{\Omega}_c)(\mathsf{R}_a\mathbf{T}_b + \mathbf{T}_a)\,]. \tag{81}$$

2) Composing a finite with an infinitesimal motion: The addition formula (81) can be analyzed in the limit $t \to 0$ where the right composition is $\ln(\mathsf{R}_b, \mathbf{T}_b) = [t\boldsymbol{\omega}_b, t\mathbf{v}_b]$. It is easy to show that the following must hold to order $\mathcal{O}(t)$,

$$\begin{aligned}&[\mathbf{\Omega}_c, \mathbf{t}_c] = [\mathbf{\Omega}_a, \mathbf{t}_a] + \\ &\quad[\mathsf{J}_{\omega_r}(\mathbf{\Omega}_a)\boldsymbol{\omega}_b,\; \mathsf{S}^{-1}(\mathbf{\Omega}_a)\mathsf{R}(\mathbf{\Omega}_a)\mathbf{v}_b + \mathsf{J}_{t_r}(\mathbf{\Omega}_a, \mathbf{t}_a)\boldsymbol{\omega}_b].\end{aligned} \tag{82}$$

3) The S^{-1} identity and coupling Jacobian: In (51) we only gave the expression for S. We can use the properties of the finite H algebra with its characteristic polynomial $\mathsf{H}(\mathsf{H}^2 + \mathsf{I}) = 0$ to obtain an expression for $\mathsf{S}^{-1}(\mathbf{\Omega}_a)$. As an interesting fact, we meet the Rodrigues Jacobians again

$$\begin{aligned}\mathsf{S}^{-1}(\mathbf{\Omega}_a) &= (1-x)\mathsf{I} + x\,\mathbf{r}_a\mathbf{r}_a^T - \frac{1}{2}[\mathbf{\Omega}_a]_\times \\ &= \mathsf{J}_{\omega_r}(-\mathbf{\Omega}_a) = \mathsf{J}_{\omega_l}(\mathbf{\Omega}_a)\,.\end{aligned} \tag{83}$$

Further, one can show that

$$\mathsf{S}^{-1}(\mathbf{\Omega}_a)\mathsf{R}(\mathbf{\Omega}_a) = \mathsf{R}(\mathbf{\Omega}_a)\mathsf{S}^{-1}(\mathbf{\Omega}_a) = \mathsf{J}_{\omega_r}(\mathbf{\Omega}_a), \tag{84}$$

A more detailed calculation for the coupling Jacobian J_{t_r} yields

$$\mathsf{J}_t(\mathbf{\Omega}_a, \mathbf{t}_a) = \frac{\partial[\mathsf{S}^{-1}(\mathbf{\Omega}_a)\mathbf{T}_a]}{\partial\mathbf{\Omega}_a}\,\mathsf{J}_{\omega_r}(\mathbf{\Omega}_a), \quad \mathbf{T}_a = \mathsf{S}(\mathbf{\Omega}_a)\,\mathbf{t}_a. \tag{85}$$

We can show that

$$\frac{\partial[\mathsf{S}^{-1}(\mathbf{\Omega}_a)\mathbf{T}_a]}{\partial\mathbf{\Omega}_a} = \frac{1}{2}[\mathbf{t}_a]_\times + \frac{x}{\Theta_a}\left(\mathbf{r}_a\mathbf{t}_a^T - t_a\,\mathsf{H}_a^2 + \mathbf{t}_a\mathbf{r}_a^T\right) + \frac{t_a}{\Theta_a}(\mathsf{S}(\mathbf{\Omega}_a) - \mathsf{S}^{-1}(-\mathbf{\Omega}_a)), \quad (86)$$

with $t_a = \mathbf{r}_a \cdot \mathbf{t}_a$, which finishes the discourse on the $SE(3)$ composition Jacobians.

F. Additive versus composite derivative

The Rodrigues formula (51) allows differentiation with respect to $\mathbf{\Omega}$ directly,

$$\mathsf{R}(\mathbf{\Omega}_0 + \Delta\boldsymbol{\omega}) \approx \mathsf{R}(\mathbf{\Omega}_0) + \left.\frac{\partial\mathsf{R}}{\partial\Omega_\mu}\right|_{\mathbf{\Omega}_0} \Delta\omega_\mu = \mathsf{R}(\mathbf{\Omega}_0)(\mathsf{I} + [\delta\mathbf{\Omega}]_\times). \quad (87)$$

We can call $\partial\mathsf{R}/\partial\Omega_\mu$ the additive derivative, whereas $[\delta\mathbf{\Omega}]_\times = \frac{d}{dt}\big|_{t=0}\exp(t[\delta\mathbf{\Omega}]_\times)$ can be termed the compositive (incremental) derivative.

In accordance with the vector addition over $SO(3)$ (67), we can show that

$$\delta\mathbf{\Omega} = \mathsf{J}_{\omega_r}^{-1}(\mathbf{\Omega}_0)\,\Delta\boldsymbol{\omega} \quad (88)$$

or explicitly

$$\delta\mathbf{\Omega} = \frac{\sin\Theta_0}{\Theta_0}\Delta\boldsymbol{\omega} + \frac{1-\cos\Theta_0}{\Theta_0}\,\Delta\boldsymbol{\omega}\times\mathbf{r}_0 + k\,(1 - \frac{\sin\Theta_0}{\Theta_0})\,\mathbf{r}_0$$

with $k = \mathbf{r}_0 \cdot \Delta\boldsymbol{\omega}$ and the usual definition $\Theta_0 = |\mathbf{\Omega}_0|$, $\mathbf{r}_0 = \mathbf{\Omega}_0/\Theta_0$.

G. Definitions in the update formula

$$\mathsf{J}_{\text{obs}} = \begin{pmatrix} {\mathsf{J}_\times}^T\mathsf{R}_{f_t}^T\mathsf{H}_t^T\Sigma_{zz}^{-T} \\ \mathsf{R}_{f_t}^T\mathsf{H}_t^T\Sigma_{zz}^{-T} \end{pmatrix}, \quad (89)$$

$$\begin{aligned} \mathsf{JMJ}(f_t,\mathbf{x}_t) &= {\mathsf{J}_\times}^T(\mathbf{x}_t)\mathsf{M}(f_t,\mathbf{x}_t)\mathsf{J}_\times(\mathbf{x}_t), \\ \mathsf{MJ}(f_t,\mathbf{x}_t) &= \mathsf{M}(f_t,\mathbf{x}_t)\mathsf{J}_\times(\mathbf{x}_t) = [\mathsf{JM}(f_t,\mathbf{x}_t)]^T. \end{aligned}$$

Further we define,

$$\mathsf{M}(f_t,\mathbf{x}_t) = [\mathsf{R}_{f_t}^T\mathsf{H}_t^T\,\Sigma_{zz}^{-1}\,\mathsf{H}_t\mathsf{R}_{f_t}]_{3\times3}\,. \quad (90)$$

H. bundle-adjustment matrix

The structure of the measurement matrix in the example depicted in Fig.5 for 3 landmarks and 2 cameras is $\Sigma_{\text{msr}}^{-1} =$

	f^3	f^2	$\mathbf{x}^1$	$\mathbf{x}2$	$\mathbf{x}^3$
f^3	$\mathsf{c}^{3,j}\mathsf{JMJ}(f^3,\mathbf{x}^j)$			$\mathsf{JM}(f^3,\mathbf{x}^2)$	$\mathsf{JM}(f^3,\mathbf{x}^3)$
f^2		$\mathsf{c}^{2,j}\mathsf{JMJ}(f^2,\mathbf{x}^j)$	$\mathsf{JM}(f^2,\mathbf{x}^1)$	$\mathsf{JM}(f^2,\mathbf{x}^2)$	
$\mathbf{x}^1$		$\mathsf{MJ}(f^2,\mathbf{x}^1)$	$\mathsf{c}^{i,1}\mathsf{M}(f^i,\mathbf{x}^1)$		
$\mathbf{x}^2$	$\mathsf{MJ}(f^3,\mathbf{x}^2)$	$\mathsf{MJ}(f^2,\mathbf{x}^2)$		$\mathsf{c}^{i,2}\mathsf{M}(f^i,\mathbf{x}^2)$	
$\mathbf{x}^3$	$\mathsf{MJ}(f^3,\mathbf{x}^3)$				$\mathsf{c}^{i,3}\mathsf{M}(f^i,\mathbf{x}^3)$

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